

Eventual ultraboundedness has a sharp nonlocal threshold

Candidate sharp partial obstruction for arXiv:2002.09221

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Status. Candidate sharp partial obstruction, likely valid. We give an exact counterexample to the broad reversible-Markov implication suggested by the source: a kernel can be bounded at one positive time without being bounded at all positive times. The example is nonlocal, so the source’s question for smooth Langevin diffusions on Euclidean space remains open here.

1 Source and exact signal

The source is P. Cattiaux, M. Fathi, and A. Guillin, *Self-improvement of the Bakry–Emery criterion for Poincaré inequalities and Wasserstein contraction using variable curvature bounds*, arXiv:2002.09221 [1]. On page 26, in the setting of the reversible diffusion generated by $L = \Delta - \nabla V \cdot \nabla$ on $\mathbb{R}^n$, the authors write:

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Actually, we do not know examples where the semi-group is not ultra bounded, but $r(2\eta, x, x)$ is bounded in x . In the ultra-bounded situation, one may directly control the W_1 decay and see that the rate of the exponential decay obtained in the previous two subsections is not optimal.

Figure 1: The exact open signal on page 26 of the source PDF.

Transcription. “Actually, we do not know examples where the semi-group is not ultra bounded, but $r(2\eta, x, x)$ is bounded in x .” Here $r(t, x, y)$ is the transition density with respect to the invariant probability μ ; “ultra bounded” means $\sup_{x,y} r(t, x, y) < \infty$ for every $t > 0$.

2 The exact reduction

Lemma 1 (one-time diagonal bound equals eventual ultraboundedness). *Let $(P_t)_{t \geq 0}$ be a symmetric Markov semigroup with densities r_t with respect to a probability measure μ . If*

$$M_T := \sup_x r_T(x, x) < \infty$$

for one $T > 0$, then $\sup_{x,y} r_s(x, y) \leq M_T$ for every $s \geq T$.

Proof. Symmetry and the semigroup identity give

$$r_T(x, y) = \int r_{T/2}(x, z) r_{T/2}(y, z) d\mu(z).$$

Cauchy–Schwarz therefore yields $r_T(x, y)^2 \leq r_T(x, x)r_T(y, y) \leq M_T^2$. For $s \geq T$,

$$r_s(x, y) = \int r_T(x, z)r_{s-T}(z, y) d\mu(z) \leq M_T,$$

because the second factor integrates to one. $\square$

Thus the source is asking whether *eventual* ultraboundedness can occur without *immediate* ultraboundedness in its local diffusion class.

3 Proof intuition

Subordination slows the high-frequency damping of a heat semigroup. On the torus, Brownian eigenvalues grow like $|k|^2$. Replacing them by $\log(1+|k|^2)$ changes exponential damping into the polynomial multiplier $(1+|k|^2)^{-t}$. This multiplier is summable only after the Sobolev threshold $2t > d$. Gamma subordination preserves positivity, constants, symmetry, and the Markov property, so every probabilistic feature except locality survives. A periodized-Gaussian lower bound proves that the failure at and below the threshold is genuine kernel blow-up, not merely failure of an absolute-convergence test.

4 The sharp threshold example

Let $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ have normalized Haar measure, let $A = -\Delta \geq 0$, and define

$$P_t = e^{-t \log(I+A)} = (I - \Delta)^{-t}, \quad t \geq 0.$$

Theorem 2 (exact eventual-ultraboundedness threshold). *The family $(P_t)_{t \geq 0}$ is a conservative symmetric Markov semigroup on $\mathbb{T}^d$. Its transition kernel is bounded if and only if*

$$t > \frac{d}{2}.$$

Consequently, for every $\eta > d/4$ one has $\sup_x r_{2\eta}(x, x) < \infty$, but the semigroup is not ultrabounded in the all-positive-times sense used in arXiv:2002.09221.

Proof. For $t > 0$ and $\lambda \geq 0$, the Gamma integral gives

$$(1 + \lambda)^{-t} = \frac{1}{\Gamma(t)} \int_0^\infty s^{t-1} e^{-s} e^{-s\lambda} ds. \quad (4.1)$$

Hence

$$P_t = \frac{1}{\Gamma(t)} \int_0^\infty s^{t-1} e^{-s} e^{s\Delta} ds. \quad (4.2)$$

This is a probability mixture of the Brownian heat operators. It is therefore positivity preserving and fixes constants. Self-adjointness is clear, and the spectral multipliers show $P_t P_u = P_{t+u}$.

The Fourier kernel is

$$K_t(z) = \sum_{k \in \mathbb{Z}^d} (1 + 4\pi^2 |k|^2)^{-t} e^{2\pi i k \cdot z}. \quad (4.3)$$

If $t > d/2$, the coefficients belong to $\ell^1(\mathbb{Z}^d)$, so (4.3) converges absolutely and uniformly to a bounded continuous kernel.

It remains to prove necessity. Let h_s denote the torus heat kernel. On the representative cube $[-1/2, 1/2]^d$,

$$h_s(z) = (4\pi s)^{-d/2} \sum_{m \in \mathbb{Z}^d} \exp\left(-\frac{|z + m|^2}{4s}\right).$$

Combining this with (4.2), retaining only $m = 0$, and taking $0 < |z| \leq 1/4$, we obtain

$$\begin{aligned} K_t(z) &\geq \frac{1}{\Gamma(t)} \int_{|z|^2}^1 s^{t-1} e^{-s} (4\pi s)^{-d/2} \exp\left(-\frac{|z|^2}{4s}\right) ds \\ &\geq c_{d,t} \int_{|z|^2}^1 s^{t-1-d/2} ds. \end{aligned} \tag{4.4}$$

When $t < d/2$, the last expression grows like $|z|^{2t-d}$; when $t = d/2$, it grows like $\log(1/|z|)$. Thus K_t is unbounded on every neighborhood of zero for all $t \leq d/2$. This proves the exact threshold. Choosing $2\eta > d/2$ gives the claimed diagonal bound, while any $0 < t \leq d/2$ disproves all-time ultraboundedness. $\square$

5 What the example settles—and what it does not

Theorem 2 shows that no abstract argument using only symmetry, reversibility, the Markov property, compact state space, or a probability invariant measure can propagate a one-time kernel bound backwards to every positive time. General eventual-ultracontractivity mechanisms under subordination are established in [2, 3]; the exact torus threshold is included here with a self-contained proof. Novelty of that specialization is not asserted.

The missing source hypothesis is locality. The generator $-\log(I - \Delta)$ is nonlocal (the process is Gamma-subordinated Brownian motion), whereas arXiv:2002.09221 works with the second-order diffusion $\Delta - \nabla V \cdot \nabla$ on $\mathbb{R}^n$. Therefore this packet is a sharp partial obstruction, not a full counterexample to the source-context question.

Eight focused upgrades were attempted. Besides the torus construction, an exact reversible star chain gives the same threshold phenomenon. A ground-state-transform search found nonlocal but no decisive local examples. A one-dimensional hitting-time route encountered an explicit correction in the authors' later paper [4]: the earlier bounded-Lyapunov implication requires extra assumptions. Standard quadratic and superquadratic potentials exhibit an all-times dichotomy. Finally, narrow remote plateaux can reproduce the necessary local lower bounds, but a uniform large-time heat-kernel upper bound across all connecting regions remains unproved.

6 Verification, novelty bounds, and review recommendation

The proof has two independent halves: ℓ^1 Fourier summability above the threshold and a positive heat-kernel lower bound below it. The latter also checks the endpoint $t = d/2$, where the singularity is logarithmic. The included verifier report records these checks and the scope boundary.

The run's four lightweight indexes were searched for arXiv:2002.09221, eventual ultracontractivity, Gamma subordination, and the core kernel phrase; no duplicate packet was found. Bounded external searches through 13 August 2026 included the exact quoted sentence, eventual/asymptotic ultracontractivity, classical Schrödinger variants, and bounded-Lyapunov criteria. They found the general subordinated-semigroup literature [2, 3] and the correction [4], but no resolution of the smooth finite-dimensional Langevin case.

Human review should check (i) that the intended scope of the source sentence indeed retains its surrounding Langevin assumptions, (ii) the endpoint in (4.4), and (iii) the classification as a partial obstruction. Promotion to a full answer requires either a smooth local diffusion with this threshold behavior or a proof that eventual ultraboundedness implies immediate ultraboundedness for all such diffusions.

References

- [1] P. Cattiaux, M. Fathi, and A. Guillin, *Self-improvement of the Bakry–Emery criterion for Poincaré inequalities and Wasserstein contraction using variable curvature bounds*, arXiv:2002.09221.
- [2] A. Bendikov, T. Coulhon, and L. Saloff-Coste, *Ultracontractivity and embedding into L^∞* , Math. Ann. 337 (2007), 817–853.
- [3] I. Gentil and P. Maheux, *Super-Poincaré and Nash-type inequalities for subordinated semigroups*, arXiv:1105.3095.
- [4] P. Cattiaux and A. Guillin, *Hitting times, functional inequalities, Lyapunov conditions and uniform ergodicity*, J. Funct. Anal. 272 (2017), 2361–2391; arXiv:1604.06336.